

ABDUL KARIM FARRAH

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About me

Software Engineer with **3 years of experience** developing production **C++** software in **Linux** environments, specializing in **high-throughput, concurrent and data-intensive systems**.

Experienced in **distributed communication, databases, messaging, network services** and internal C++ frameworks, with expertise in **performance optimization, production debugging, large-scale data processing, testing, migration and production support**.

Technical Skills

Programming Languages:

C++, C, Python, SQL, Java

C++ & Software Engineering:

C++11, C++14, C++17, C++20, Modern C++, STL, Object-Oriented Programming (OOP), Templates, Generic Programming, Smart Pointers, RAII, Move Semantics, Lambdas, Iterators, STL Algorithms, Data Structures & Algorithms, Resource Management, Exception Handling

Concurrency & Performance:

Multithreading, Concurrency, Synchronization, Thread Safety, Asynchronous Processing, Thread Pools, Mutexes, Condition Variables, Semaphores, Atomics, Read/Write Locks, Lock-Free Data Structures, Race Conditions, Deadlock Debugging, High-Throughput Processing, Low-Latency Processing, Performance Optimization, Performance Tuning, Benchmarking, Load Testing, Stress Testing

Linux & Systems:

Linux, Unix/Linux, POSIX, Linux System Programming, IPC, Process/Thread Management, Socket Programming, TCP/IP, Network Programming, Bash, Shell Scripting, GDB, Valgrind, Core Dumps, Stack Trace Analysis, Memory Leak Detection, Production Debugging

Distributed Systems & Communication:

gRPC, Protobuf, TCP, HTTP, HTTP/2, REST APIs, Kafka, Event-Driven Processing, Asynchronous Communication, Message Processing, Cache Synchronization, Rate Limiting, JSON, XML

Databases & Data Engineering:

SQL, MySQL, PostgreSQL, Database Design, Schema Design, Indexing, Composite Indexes, Query Optimization, EXPLAIN, Partitioning, Transactions, Prepared Statements, Connection Pooling, Bulk Loading, ETL, Data Migration, Data Transformation, Large-Scale Data Processing

Testing & Quality:

GoogleTest, Unit Testing, Functional Testing, Integration Testing, Performance Testing, Load Testing, Stress Testing, Test Automation, Validation, Regression Testing

Build, Version Control & DevOps:

GCC, G++, Make, Makefiles, CMake, Git, GitLab, GitLab CI/CD, Build Automation, Release/Artifact Management, Docker

Additional Technologies:

MemCache, TinyXML, Oracle Wrappers, FastAPI, Django, React Native, HTML, CSS, JavaScript

Work experience

Software Engineer | 6D Technologies | 02/01/2024 - Current | Bengaluru, India

Develop and maintain production software using internal **C++ frameworks, database wrappers, logging components, threading utilities, thread-safe containers and custom data structures**, across high-throughput processing, telecom/network infrastructure, database migration, distributed messaging and production troubleshooting.

Projects

C++ Frameworks, Systems Development & Engineering

- Develop production applications primarily in **C++11/14/17** on Linux using internally developed C++ frameworks and wrappers for database access, STL utilities, logging, threading, thread-safe containers and communication.
- Rapidly learn and apply internal frameworks, wrappers and custom data structures while maintaining compatibility with established production architecture.
- Extend, debug and validate internal framework components when application requirements or framework-version changes expose implementation issues.

- Apply **OOP, STL containers and algorithms, templates, smart pointers, RAII, move semantics and lambdas** to develop maintainable application components.
- Select data structures based on workload, lookup patterns and concurrency requirements, using STL containers and custom structures for routing and cache-processing logic.
- Develop concurrent processing logic using **threads, mutexes, condition variables, semaphores, atomics, read/write synchronization and thread-safe containers**.
- Diagnose and resolve **race conditions, synchronization issues and deadlocks** using GDB stack traces, thread information, logs and production diagnostics.
- Use **GDB, Valgrind, core dumps and log analysis** to troubleshoot crashes, memory leaks, thread behavior and runtime issues.
- Work extensively in Linux using **Bash, grep, awk, sed, find, xargs, nohup and system/service utilities** for development, testing, troubleshooting and automation.
- Use Git/GitLab for source control and participate in **GitLab CI/CD, tagging and application artifact release workflows**.

High-Throughput Call Processing & Distributed Messaging

- **Project Scope:** Production C++ applications responsible for post-call message processing and downstream communication across multiple deployments.
- Developed and maintained high-throughput **C++ applications** processing post-call requests through **gRPC/Protobuf** interfaces.
- Implemented request parsing, validation and business-condition processing before forwarding requests through **TCP or gRPC communication wrappers**.
- Developed asynchronous/non-waiting processing logic, allowing concurrent request handling without unnecessarily blocking processing threads.
- Contributed to concurrency and throughput optimization; redesign of asynchronous pool logic resulted in approximately **5× improvement in observed TPS**.
- Implemented and maintained continuously running **HTTP authentication servers** for authorization requests.
- Integrated **CDR generation/writing modules** for transaction and billing-related information.
- Used **Kafka-based cache synchronization** to maintain application state across instances and support distributed/load-balanced deployments.
- Used MySQL-backed state and reload mechanisms to restore application information and periodically refresh caches.
- Implemented per-MSISDN message limits and business rules controlling daily successful and promotional message processing.
- Worked with **multithreaded cache and rate-limiting logic** where concurrent requests updated shared counters.
- Redesigned counter-handling logic to prevent multiple threads from simultaneously entering a limited processing path after threshold checks.
- Used **MemCache** for shared success-message counting and rate-limiting in a highly concurrent application.
- Implemented an internal **gRPC communication path** in one deployment to replace a database lookup, reducing processing overhead.
- Worked with multiple **Protobuf definitions and gRPC request/response structures** for different communication requirements.

Large-Scale Data Migration & ETL — TFN Routing Systems

- **Project Scope:** Migration and transformation of legacy routing datasets into optimized MySQL-backed application data structures.
- Analyzed legacy routing data and business behavior, including **time-based, sequential, direct and logical routing mechanisms**, to understand the existing platform representation.
- Cleaned, transformed and processed large legacy datasets to match target data models and operational requirements.
- Loaded processed source data into MySQL and developed **C++ migration applications** to read source databases, transform routing information and populate target schemas.
- Designed efficient data structures for **finding, deleting, inserting and caching routing information** based on routing requirements.
- Optimized source database access through **indexing and hash-based lookup strategies**, substantially reducing database lookup time.
- Worked with datasets containing **tens of millions of records**, with attention to memory usage, database access patterns and processing throughput.
- Used C++ and Python tooling for data transformation, validation, ETL processing and migration verification.
- Applied SQL query optimization and database indexing to reduce long-running migration operations from **hours to minutes** in relevant workloads.

Large-Scale XML / TFN Routing Migration

- Analyzed legacy routing data containing **time-based, sequential, percentage-based and logical-destination routing configurations**.
- Processed large XML-based datasets using **TinyXML** to parse and transform legacy routing information.
- Developed C++ migration applications to read transformed data from MySQL source tables, apply routing-specific processing and populate optimized target schemas.
- Designed custom data structures for application cache processing based on routing requirements and lookup patterns.
- Optimized database access using **indexes and hash-based lookup strategies** for migration and transformation workloads.
- Developed a C++ application to load **Keycloak-related configuration/details** into the target environment for subsequent HTTP-based access.

5G Equipment Identity & Network Services

- **Project Scope:** Equipment Identity Register and 5G Service-Based Architecture applications.
- Worked on **5G Equipment Identity Register (5G-EIR)** systems used for equipment identity verification during network procedures.
- Studied and implemented requirements based on relevant **3GPP architecture and service-based interface concepts**.
- Worked with the **N5g-eirEquipmentIdentityCheck** network function used by the AMF to determine whether a device equipment identity is present in a prohibited list.
- Implemented HTTP-based service interfaces for equipment-status requests containing **PEI and optional SUPI/GPSI information**.
- Worked with **REST-style HTTP APIs, JSON and ProblemDetails responses**, including success and 4xx/5xx error handling.

- Worked with HTTP/2-based Service-Based Interface communication involving **IP, TCP, TLS and HTTP/2**
- Worked with authorization concepts involving **OAuth 2.0**, NRF-provided access tokens and network-function service authorization.
- Gained practical exposure to **SEPP-mediated communication, service discovery and redirection between 5G-EIR service instances**.

Centralized Equipment Identity Management

- Developed and maintained components of a centralized **Equipment Identity Register** application handling **IMEI/IMSI/MSISDN triplet data**
- Implemented initial database loading and application-side processing of large MySQL datasets.
- Developed communication flows using **gRPC and Kafka** for adding, updating and deleting triplet information.
- Used Kafka for cache synchronization and downstream message propagation between application components.
- Implemented processing logic that transforms received gRPC requests into required downstream Kafka events.
- Worked with database-backed application state and distributed cache synchronization in a production-oriented environment.

MySQL Framework Simulator & Database Infrastructure Testing

- Developed and tested an internal **MySQL wrapper/testing module** covering direct and prepared queries with different parameter types, including strings, integer/long values and date/time data.
- Created test applications to validate database-wrapper behavior and exercise different SQL execution paths.
- Performed **load testing** to evaluate database-module throughput and behavior under sustained workloads.
- Tested framework-version changes and investigated/fixed errors introduced by internal framework modifications.
- Diagnosed issues involving **prepared statements, SSO-related database interactions and query execution**.
- Validated an internal MySQL library during a framework-version upgrade to ensure existing application use cases continued to work correctly.

Production Kafka Incident Resolution — Cache Cleanup

- Worked on a production cache-cleanup application responsible for removing obsolete **DID/MSISDN records** from application caches and MySQL-backed state.
- Investigated a production issue where DID numbers were not removed because Kafka messages were generated using an incorrect packet format.
- Developed a targeted **C++ remediation application** to retrieve affected DID numbers from the database and generate the required Kafka messages.
- Designed the fix to process affected records within the required operational window while minimizing impact on existing application traffic.
- Created a **Python + ADB validation script** to automatically place calls to affected numbers and verify expected call-disconnection behavior after remediation.
- Used production logs, database state and message-processing behavior to isolate the issue and validate the corrective action.

Database Churn & High-Volume Data Processing

- Developed a C++ application to remove unwanted records from large database tables while reducing the performance impact of conventional row-by-row deletion.
- Redesigned the target table schema to support partition-based data management while preserving legacy column names and application compatibility.
- Developed a **Bash automation script** to orchestrate the database-processing workflow.
- Designed a dump-based approach that extracted database contents, modified required SQL data at file level and loaded the resulting dataset into the target table instead of relying on large in-place DELETE operations.
- Worked with a complete source dataset of approximately **95 million records**.
- Applied file-based transformation, database loading and validation techniques to reduce processing time compared with conventional deletion of very large datasets.
- Used C++ and Python tooling to process, validate and transform large data volumes.

XML Report Generation & Scheduled Processing

- Developed a C++ **XML report-generation application** that retrieves data and report configuration from MySQL and generates XML reports according to configurable formats.
- Combined multiple source formats into the required output representation before generating report files.
- Implemented scheduled report generation where the application performs its work and then enters a sleep state until the next execution.
- Designed the processing flow to reduce unnecessary CPU consumption while maintaining reliable wake-up and report generation.

Additional Engineering Experience

- Worked with **Docker images** for compiling and executing applications inside controlled development environments.
- Used internal **Oracle database wrappers** for application-level database access.
- Developed C++ and Python utilities for testing, validation, data processing and reproducing framework/library behavior.
- Assisted teammates with application-level debugging, framework usage and technical problem solving while owning assigned modules and implementation areas.

Education & Training

Bachelor of Technology | APJ Abdul Kalam Technological University | 03/06/2019 - 23/08/2023 | Thiruvananthapuram, India

Field(s) of study: Computer Science and Engineering | **Final grade:** 8.95 / 10 | **Level in EQF:** 6

Technical Projects

High-Performance C++20 HFT Matching Engine

- **C++20 | Multithreading | STL | CMake | GoogleTest | Data Structures | Performance Benchmarking**
- Designed and developed a production-oriented **C++20 matching engine** supporting multiple independent symbols and order books using deterministic **Price-Time Priority** matching.
- Implemented **Limit, Market, IOC, Cancel and Modify** orders, including partial, complete and multiple fills.
- Designed a concurrent architecture with a dedicated worker thread and request queue per symbol, enabling **parallel processing across symbols while preserving deterministic sequential processing within each symbol**.
- Implemented thread-safe shared components including logging and market-data publication using synchronization primitives.
- Used std::map, std::list, std::unordered_map, std::queue and std::unordered_set for order-book operations and order lookup.
- Implemented duplicate-order detection, validation, error handling and self-trade prevention.
- Developed market-data publication, execution logging and **snapshot/recovery** functionality for application state persistence.
- Implemented JSON-based state persistence and recovery.
- Benchmarked throughput and latency using order-processing workloads and measured **mean, P50, P75, P90, P99 and maximum latency**.
- Achieved approximately **56,978 orders/second** in one 100,000-order benchmark workload, with performance dependent on the execution environment.
- Developed **43 GoogleTest unit tests** covering matching logic, order operations, validation and system behavior.
- Used **CMake** for project configuration and build management.

SysDash — Linux System Resource Monitor

- **C++ | Linux | /proc | /sys | ncurses**
- Developed a terminal-based **Linux system monitoring application in C++** for displaying system resource information.
- Integrated Linux /proc and /sys interfaces to obtain runtime system information.
- Built an interactive terminal UI using **ncursesw**.
- Worked directly with Linux system-level data and terminal-based resource monitoring.

Pokémon 2D Battle Simulator

- **Python | Pygame | Networking / Host Logic | Game State Management**
- Developed a 2D Pokémon battle simulator using **Python and Pygame**.
- Implemented the battle engine, game-state handling, player interactions and visual rendering.
- Designed host/room logic to manage multiplayer-style session flow and game-state coordination.
- Built the application architecture and gameplay logic independently, including state transitions and user interaction handling.

Local AI Workbench

- **Python | FastAPI | Mistral 7B | Stable Diffusion | HTML | CSS | JavaScript**
- Developed an offline AI workbench combining **local large-language-model inference and image generation** into a single application.
- Integrated a locally hosted **Mistral 7B** model for conversational AI functionality.
- Built a FastAPI backend to expose application services and coordinate AI inference workflows.
- Integrated Stable Diffusion-based image generation into the application.
- Developed the web interface using **HTML, CSS and JavaScript**.
- Implemented session/history handling for maintaining previous application interactions.
- Designed the system to operate locally without relying on external AI inference services.